Set Theoretic Estimation, Recast in Cohomology

A standalone, low-jargon note

Project `thejeb`

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Abstract

This note explains, in as little jargon as possible, one idea: the gap between *local agreement* and a *global solution* in set theoretic estimation (STE) is exactly the kind of gap that cohomology was invented to measure. We define every term as it appears, work one tiny example by hand, and give a plain dictionary between the STE words and the sheaf/cohomology words. Everything asserted as a theorem here is machine-checked in the project's Lean library and cited by name; the rest is clearly flagged as intuition. No prior cohomology is assumed.

1 STE in one paragraph

Set theoretic estimation [4] models a problem as a collection of *property sets*. Each source of information — a sensor, a document, a voice — contributes a set S_i of the candidate answers consistent with that source. The *feasible set* is the intersection

$$F = \bigcap_i S_i,$$

the answers consistent with *everyone at once*. The corpus is *feasible* when $F \neq \varnothing$ and *infeasible* when $F = \varnothing$. That is the whole object of study: does a single answer satisfy every source, and if so, which ones?

2 The one idea

Checking $F \neq \varnothing$ directly means reasoning about all sources and all variables simultaneously — expensive. In practice we would rather check things *locally*: does each variable, looked at on its own, have a consistent value? Do each small group of sources agree on the piece they share?

The catch is that local agreement need not add up to a global answer. It is possible for every small piece to be internally consistent, and for every pair of overlapping pieces to agree where they overlap, and yet for *no* single global assignment to restrict to all of them at once. The pieces fit pairwise but cannot be assembled.

> **The recast.** The obstruction to assembling locally-agreeing pieces into one global solution is a *cohomology class*. When it vanishes, local agreement guarantees a global answer, and STE becomes a local computation. When it does not vanish, it measures exactly how much the problem is genuinely coupled — how far local reasoning falls short.

This is the same phenomenon cohomology measures everywhere it appears: a consistent choice made on every small patch that provably cannot come from one choice made on the whole. We now make each word precise.

3 The building blocks, defined plainly

Fix a set V of *variables*. Each variable v ranges over some value set A_v.

Global configuration. One choice of a value for every variable at once: a function f with $f(v) \in A_v$ for all v. Write the set of all of them as $\prod_v A_v$.

Constraint. A set T of allowed global configurations, $T \subseteq \prod_v A_v$. In STE, T is the feasible set F; the recast just insists on viewing F as a subset of the full configuration space, so we can talk about restricting it.

Context (scope). A subset $U \subseteq V$ of variables — a "window" we are willing to look at all at once. A small context is cheap; the full set V is the expensive global view.

Cover. A collection of contexts $\{U_j\}$ that together touch every variable (every v lies in some U_j). This is how we chop the global problem into local windows.

Local section. A partial assignment on a context U_j — a choice of value for just the variables in that window — that *could* be extended to a full allowed configuration in T. "A local piece that is not obviously doomed."

4 Local agreement versus a global solution

Two definitions carry the whole idea.

Compatible family (local agreement). Pick one local section s_j on each context U_j, such that any two of them *agree on the variables their contexts share*:

$$\text{for all } j, k \text{ and every } v \in U_j \cap U_k, \qquad s_j(v) = s_k(v).$$

This is "everything is locally fine and the overlaps match." It is the cheap, checkable condition. (In cohomology this overlap-agreement is called the *cocycle condition*.)

Gluing (a global solution). The family *glues* if there is a single global configuration $f \in T$ whose restriction to each window U_j is exactly s_j. One answer that explains all the local pieces simultaneously. This is the expensive thing we actually want.

The obstruction. Every genuine global solution obviously produces a compatible family (just restrict it to each window). The question is the converse:

Does every compatible family glue?

When the answer is *yes* for a constraint T and a cover, we say T **glues over that cover**, or that its *Čech obstruction vanishes*. Local agreement then *is* global solvability. When the answer is *no*, some compatible family is locally perfect but globally impossible; the number of such stuck families is the size of the obstruction — the cohomology. A vanishing obstruction is the statement "$H^1 = 0$" in this setting; a nonzero one is a nonzero H^1 class.

5 A dictionary

STE / plain word	Sheaf-cohomology word	What it means
Variable value space A_v	Stalk / local data	values a coordinate can take
Constraint / feasible set T	The (sub)presheaf of sections	the allowed answers
Context U_j	Open set / patch	a window of variables
Partial answer on U_j	Local section	a value on that window
Extendable partial answer	Section of the (sub)sheaf	a locally-valid piece
Overlaps match	Cocycle / compatibility	pieces agree where they share
One answer fits all pieces	Global section / gluing	a genuine global solution
Local agreement $\Rightarrow$ global	Sheaf condition / $H^1 = 0$	the obstruction vanishes
Stuck locally-fine families	Nonzero H^1 class	the coupling obstruction

6 One example, by hand

Take two variables x, y, each Boolean ($A_x = A_y = \{0, 1\}$), and the cover by the two single-variable windows $\{x\}$ and $\{y\}$. There is no overlap, so "overlaps match" is automatic — *every* combination of a local value for x and a local value for y is a compatible family. There are $2 \times 2 = 4$ of them.

A gluable constraint (uncoupled). Let T be the "rectangle" $\{0, 1\} \times \{0, 1\}$: anything goes. All four compatible families glue — each is just a global configuration read locally. Obstruction: zero. More generally any *product* constraint $\prod_v P_v$ (each variable independently confined to some allowed set P_v, no cross-variable link) glues over *every* cover.

A non-gluable constraint (coupled). Let T be the "diagonal" $\{(0, 0), (1, 1)\}$: the two variables are forced equal (this is the XOR/equality link). Locally each variable can still be 0 or 1, so all four compatible families are still there. But only *two* of them — $(0, 0)$ and $(1, 1)$ — come from a global configuration in T. The mixed family "$x \mapsto 0,\ y \mapsto 1$" is locally flawless and globally impossible. Two stuck families remain: the obstruction is $2 \neq 0$. The coupling between x and y is invisible in any single window and shows up *only* as this obstruction.

That contrast — rectangle glues, diagonal does not — is the entire mechanism in miniature.

7 The dichotomy, and why it is useful

The example is not a coincidence. In the project's Lean library the two sides are theorems:

- **Uncoupled $\Rightarrow$ gluable.** Every product ("rectangular") constraint glues over every cover (`rectangular_cechVanishesCover`).
- **Gluable $\Rightarrow$ uncoupled.** If the obstruction vanishes, the constraint *must* be a rectangle (`CechVanishes` forces `IsRectangle`, in module `Ste.CechObstruction`).
- **The witness.** The diagonal has obstruction exactly 2, a genuinely nonzero class (`Ste.CechObstruction`).

So, in this setting, *gluable* = *rectangular* = *uncoupled* are one and the same, and the obstruction is a sharp detector of coupling.

Why it matters for solving problems:

1. **Gluable problems are cheap.** If the obstruction vanishes, you never assemble a global solution by search. You solve each window — in the extreme, each single variable — and local agreement is a *theorem* that the global answer exists. For the "frame" corpora the project studies (each source asserts unary `variable = value` facts), feasibility collapses to a one-variable-at-a-time scan: a value is consistent iff no two sources put different values on the same variable. This is the same fact as "pairwise agreement forces global agreement" — the Helly-number-2 property (`frame_hasHelly_two`, module `Ste.HellyConsistency`) — and it is why the project's solver certifies large corpora by a linear sweep instead of an all-subsets search.

2. **The obstruction localizes the hard part.** When it does *not* vanish, its support tells you *which* variables are genuinely entangled and must be reasoned about jointly. You pay the global price only there. Substitution-cipher key spaces (Carlson's setting, 3) are the coupled extreme: they are not rectangular, so the obstruction is essential and local reasoning cannot finish the job.

In one line: **the Čech obstruction is a computable measure of how far an STE problem is from being solvable window-by-window**, zero exactly when it factors into independent pieces.

8 What is actually machine-checked

To keep the bright line the project insists on, here is what is a theorem (checked by Lean's kernel, sorry-free) versus what is intuition.

Theorems (names are Lean identifiers on the project's main branch): the compatible-family / gluing / vanishing definitions (`CompatibleFamily`, `GluesCover`, `CechVanishesCover` in `Ste.CechCover`); rectangles glue over every cover (`rectangular_cechVanishesCover`); vanishing forces rectangularity and the diagonal's obstruction is 2 (`Ste.CechObstruction`); the support-cover form, that vanishing is equivalent to the constraint being generated by pieces whose scopes fit the cover (`cechVanishesCover_iff_exists_scopedGeneration` in `Ste.SupportCover`); and the frame Helly-2 collapse (`frame_hasHelly_two`).

Intuition / framing, not claimed as proven here: the word "cohomology" as the *full* Čech H^1 machinery — a cochain complex with a coboundary map and $H^1 = \ker / \mathrm{im}$ over a coefficient module — is the Abramsky and Brandenburger [1], Abramsky et al. [2] sheaf-theoretic framework the project draws on. This note uses the $H^1 = 0$ / obstruction-vanishing *level*, which is what is mechanized; the graded cochain-complex construction in full generality is cited, not reproved here.

The recast, restated

STE asks whether many sources share a common answer. Split the question into windows. "Every window is fine and the overlaps match" is cheap to check. Whether that local agreement assembles into one global answer is governed by a single obstruction: it vanishes exactly for the uncoupled (rectangular) problems, and then STE is a local computation; it is nonzero exactly when the problem is genuinely coupled, and then it measures and locates the coupling. That obstruction is the cohomology, and that equivalence is the recast.

References

[1] Samson Abramsky and Adam Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics*, 13:113036, 2011. doi: 10.1088/1367-2630/13/11/113036. arXiv:1102.0264.

[2] Samson Abramsky, Shane Mansfield, and Rui Soares Barbosa. The cohomology of non-locality and contextuality. *Electronic Proceedings in Theoretical Computer Science*, 95:1–14, 2012. doi: 10.4204/EPTCS.95.1. Proc. 8th International Workshop on Quantum Physics and Logic (QPL 2011). arXiv:1111.3620.

[3] Albert H. Carlson. *Set Theoretic Estimation Applied to the Information Content of Ciphers and Decryption*. Ph.d. dissertation, University of Idaho, Moscow, Idaho, USA, May 2012. Retrieved via Internet Archive snapshot (2024-07-13) of the author's ResearchGate full text, publication 349064680.

[4] Patrick L. Combettes. The foundations of set theoretic estimation. *Proceedings of the IEEE*, 81 (2):182–208, 1993. doi: 10.1109/5.214546. Retrieved from the author's page, `https://pcombet.math.ncsu.edu/proc.pdf`.